Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (EC) Examination

May 2016

EC 308 Antenna and Wave Propagation

Date: 17/05/2016 **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the below questions.** **[07]**

a. Define the following terms with respect to antenna. **[04]**

1. Radiation Intensity
2. Stray Factor
3. Resolution
4. Beam solid angle

b. Which is a non-resonant antenna? **[01]**

A. Rhombic B. Folded dipole

C. End-fire array D. Yagi-Uda

c. What is mean by antenna tapering? **[01]**

d. Which condition should be satisfied, so that circular small loop and square small loop will have same field pattern and field intensity in far field. **[01]**

Q – 2 **Answer any two questions.** **[14]**

a. Explain case of two point sources with equal amplitude and equal phase. Find out maxima, minima and HPBW points. Also draw radiation pattern for the same case. **[07]**

b. Derive the radiation resistance of small loop antenna. **[07]**

c. Write a short note on various types of long wire antenna and design a rhombic antenna for operating frequency f=18GHz and $\theta=58^o$. **[07]**

Q - 3 **Answer any two questions.** **[14]**

a. Write the equation of Scalar potential and vector potential for short dipole. Derive the equation of electric field components and magnetic field components from above. **[07]**

b. Design a 5 turn Helical Antenna which at 400 MHz operates in the normal mode, the spacing between the turns is $\lambda/50$. It is desired that antenna possess circular polarization. Determine circumference of helical antenna, length of single turn, overall length of helix, length of wire and pitch angle. **[07]**

c. Design a three element broadside array of $\lambda/2$ spacing between elements. The pattern is to be optimum with r=25 dB. **[07]**

SECTION – II

Q - 4 Answer the below questions. **[07]**

a. Choose correct option: The purpose of grounds is to........ **[01]**
A. have resistance as high as possible.
B. have conductivity as low as possible.
C. increase ground losses and to provide the best energy from the antenna.
D. introduce the least possible amount of resistance in the ground connection.

b. What is LPDA? **[01]**

c. Fill the blank: The chief factor that controls long distance communication is the _________ of the ionized layer. (location/density/size/color) **[01]**

d. Define Lens Antenna. **[01]**

e. What is Slot Antenna? **[01]**

f. Define Duct Propagation? **[01]**

g. What is Fading? **[01]**

Q - 5 Answer the below questions. **[14]**

a. What is the importance of parabolic structure as a reflector in antenna? Explain the cassegrain feed mechanism of parabolic reflector antenna. **[05]**

OR

a. Enlist the advantage and disadvantage of the Micro-strip antenna. **[05]**

b. Write a brief note on Babinet's Principal for the complementary antennas. **[04]**

c. What is log periodic antenna? Explain with neat sketch log periodic dipole array. **[05]**

OR

c. Derive the design equation of non-metallic dielectric Plano convex lens antenna. **[05]**

Q - 6 Answer the below questions. **[14]**

a. Explain the method for the impedance measurement. **[07]**

b. Explain different layers of ionosphere (Along with typical height diagram). **[05]**

c. Explain the experimental setup for the measurement of far field radiation pattern of antenna. **[02]**

OR

a. Write the fundamental equation of FRIIS formula for free space propagation and explain in brief the procedure that helical antenna generate circularly polarized wave with the experimental set up. **[05]**

b. Explain various types of corner reflector antenna. **[05]**

c. Define the following terms **[04]**

1. Skip Distance 2. MUF
3. Virtual Height 4. Multihop Propagation

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